

Geometric origin of the quantum–classical transition: Spectral attractor in an FCC lattice with zero free parameters

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Abstract

We present a unified discrete spacetime model on a three-dimensional face-centered cubic (FCC) lattice governed by the discrete nonlinear Schrödinger equation (DNLS). The model has zero free parameters: the lattice coordination number $Z = 12$ fixes the background perturbation $\varepsilon_0 = 1/12$, the nonlinearity $\gamma = 1/\varepsilon_0 = 12$, and the spectral collapse rate $\Gamma_k = \sigma(\lambda_k) \varepsilon_0 / (\varepsilon_0 + |c_k|^2)$. The Laplacian spectrum matches the Bloch-Floquet formula exactly. The spectral zeta function satisfies the Epstein functional equation for the D_3 lattice. Time reversal symmetry is broken by an irreversible spectral purification: the von Neumann entropy $S(t) \rightarrow 0$ vanishes as all stochastic realizations converge to the same pure state (rank-1 attractor). In 3D the spectral collapse yields a stable fixed point (width 2.96 sites) while in 1D the soliton degrades (width $3.77 \rightarrow 6.52$). Five falsifiable experimental predictions are given, realizable with existing cold-atom optical lattice platforms.

1 Introduction

The transition between quantum superpositions and definite classical outcomes—the measurement problem—has resisted resolution for nearly a century [1, 2]. Proposed solutions typically introduce ad-hoc parameters that undermine predictive power. Here we show that the transition emerges naturally from the geometry of a discrete lattice without any free parameters.

Our approach combines the FCC lattice (coordination $Z = 12$), the discrete nonlinear Schrödinger equation (DNLS), and a spectral self-regulation mechanism that collapses only high-frequency modes while preserving low-frequency structure. The FCC Laplacian spectrum connects naturally to Epstein’s zeta function for the D_3 lattice [3, 4], and through the Friedli–Karlsson theorem [5], to the Riemann hypothesis.

2 FCC lattice and Laplacian

The FCC lattice is defined on sites (i, j, k) with $i, j, k \in \mathbb{Z}$ and $(i + j + k)$ even, in a periodic supercell $N \times N \times N$. Each site has $Z = 12$ neighbors:

$$(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1). \quad (1)$$

The graph Laplacian $L = D - A$ has diagonal $D_{nn} = 12$ and adjacency $A_{nm} = 1$ for neighbors. Its Bloch–Floquet eigenvalues are

$$\lambda(k) = 12 - 4[\cos k_x \cos k_y + \cos k_x \cos k_z + \cos k_y \cos k_z], \quad (2)$$

with spectrum $[0, 16]$. For $N = 6$ (108 sites) numerical eigenvalues match Eq. (2) to machine precision (2.84×10^{-14}). For $N = 12$ (864 sites) diagonalization takes 0.96 s.

The spectral zeta function $\zeta_\Delta(s) = \sum_{\lambda_n \neq 0} \lambda_n^{-s}$ relates to Epstein’s zeta function for D_3 by

$$\xi(s) = \pi^{-s} \Gamma(s) \zeta_{D_3}(s), \quad \xi(3/2 - s) = \xi(s), \quad (3)$$

verified to $< 2\%$ error at $s = 3/4$ for $N = 8$ (256 sites).

3 DNLS and spectral collapse

The discrete nonlinear Schrödinger equation

$$i\frac{d\psi_n}{dt} = \sum_m L_{nm}\psi_m - \gamma|\psi_n|^2\psi_n \quad (4)$$

is solved via Strang splitting in the eigenbasis $\{\phi_k, \lambda_k\}$ of L . The nonlinear half-step is

$$\psi_n \rightarrow \psi_n e^{i\gamma|\psi_n|^2\Delta t/2}, \quad (5)$$

and the linear step is diagonal in the eigenbasis:

$$c_k \rightarrow c_k e^{-i\lambda_k\Delta t}, \quad c_k = \langle\phi_k|\psi\rangle. \quad (6)$$

The collapse acts in the spectral domain as a Kraus map:

$$\rho \rightarrow \frac{\sum_k K_k \rho K_k^\dagger}{\text{Tr}(\sum_k K_k \rho K_k^\dagger)}, \quad (7)$$

with Kraus operators

$$K_k = \sqrt{1 - \beta\sigma(\lambda_k) \frac{\varepsilon_0}{\varepsilon_0 + |c_k|^2}} |\phi_k\rangle\langle\phi_k|, \quad (8)$$

diagonal in the eigenbasis. Trace preservation is canonical; the normalization factor in Eq. (9) is its numerical implementation. On state vectors this reduces to the amplitude map

$$c_k \rightarrow c_k \left(1 - \beta\sigma(\lambda_k) \frac{\varepsilon_0}{\varepsilon_0 + |c_k|^2}\right), \quad (9)$$

with $\varepsilon_0 = 1/12$ and the smooth sigmoid filter

$$\sigma(\lambda) = \frac{1}{1 + e^{-s(\lambda/\lambda_{\max} - 0.5)}}, \quad (10)$$

with $s = 8$ and $\lambda_{\max} = 16$. Low-frequency modes ($\lambda \ll 8$) have $\sigma \approx 0$ and are protected; high-frequency modes ($\lambda \gg 8$) collapse at a rate self-regulated by $|c_k|^2$.

The parameter β is fixed by the FCC packing fraction $\eta = \pi/(3\sqrt{2}) \approx 0.74048$ and the background perturbation:

$$\beta = (1 - \eta) + \frac{\varepsilon_0}{2} = 1 - \frac{\pi}{3\sqrt{2}} + \frac{1}{24} \approx 0.30119 \approx 0.30. \quad (11)$$

The first term $(1 - \eta) = 0.2595$ is the void fraction of the densest sphere packing—the quantum space available for fluctuations. The second term $\varepsilon_0/2 = 1/24$ is half the fundamental perturbation scale. The attractor is stable for $\beta \in [0.1, 0.5]$; the geometric value $\beta = 0.30$ lies centrally within this range.

The von Neumann entropy over M stochastic realizations

$$\rho(t) = \frac{1}{M} \sum_{i=1}^M |\psi_i(t)\rangle\langle\psi_i(t)|, \quad (12)$$

$$S(t) = -\text{Tr}(\rho(t) \ln \rho(t)), \quad (13)$$

provides the arrow of time: $S(t) \rightarrow 0$ indicates purification.

The collapse couples the DNLS system to an external reservoir (the lattice degrees of freedom: phonons, vibrations). The second law requires $S_{\text{total}} = S_{\text{DNLS}} + S_{\text{reservoir}} \geq 0$; the purification $S_{\text{DNLS}} \rightarrow 0$ represents entropy transfer from the system to the reservoir. This is the standard mechanism of dissipative state preparation [10] and is consistent with the quantum Mpemba effect [7].

4 Results

4.1 Spectral fixed point in 3D FCC

For $N = 6$ (108 sites), $\gamma = 12$, $\beta = 0.3$, $\Delta t = 0.005$, and an initial Gaussian wavepacket ($\sigma = 2.0$, norm 2.0):

Table 1: Convergence to the spectral fixed point in 3D FCC.

t	width	$\max \psi ^2$	energy	$S(t)$
0.0	2.20	0.040	0.977	0.0124
0.5	2.96	0.009	-0.056	0.0070
1.0	2.96	0.009	-0.056	2×10^{-5}
1.5	2.96	0.009	-0.056	0

After $t \approx 1.5$ all observables are constant. The density matrix rank drops from 68 to 1: all $M = 100$ stochastic realizations converge to the same pure state.

4.2 Dimensional dependence

In 1D ($N = 64$ sites), spectral collapse degrades the soliton:

$$\text{width: } 3.77 \rightarrow 6.52, \quad \text{energy: } -0.43 \rightarrow -0.25, \quad (14)$$

over 2000 steps. No fixed point exists because the 1D soliton distributes power across many Fourier modes, none protected.

In 3D FCC, the uniform mode ($\lambda = 0$, $\sigma(0) \approx 0.018$) retains 84% of the power and is virtually immune to collapse.

4.3 Finite-size scaling

For $N = 12$ (864 sites), varying the initial norm ($\sigma = 2.0$):

Table 2: Self-trapping threshold in 3D FCC at $N = 12$.

Norm	width($t = 0$)	width($t = 2.5$)	State
2	2.45	7.41	Expansion
4	2.45	3.45	Self-trapped
8	2.45	3.69	Self-trapped

The self-trapping threshold lies between norm 2 and norm 4. Below threshold the wavepacket disperses; above it forms a coherent structure. The $N = 6$ result (width 3, norm 2) was a finite-size artifact.

4.4 Eigenmode structure

At the fixed point ($N = 12$, norm 2), 99.9% of power concentrates in the two lowest modes:

$$\text{mode 2 } (\lambda = 1.07) : 54.8\%, \quad (15)$$

$$\text{mode 0 } (\lambda = 0) : 45.1\%. \quad (16)$$

Spectral collapse selects the lowest-frequency modes as the “classical kernel.”

5 Global attractor theory

The combined map $F = \Psi \circ \Phi : S \rightarrow S$ (collapse Φ then DNLS Ψ) is continuous on the compact sphere S^{2N-1} . By Brouwer’s fixed-point theorem, F has at least one fixed point.

At the fixed point, the fixed-point condition $c_k^* = F(c_k^*)$ from Eq. (11) requires $\sigma(\lambda_k) \varepsilon_0 / (\varepsilon_0 + |c_k^*|^2) = 0$ for each mode k . Hence all power must reside in modes with $\lambda_k \ll 8$.

The flow of spectral power is unidirectional: collapse transfers power from high to low frequency, and the DNLS cannot reverse this. This irreversibility constitutes the arrow of time at the spectral level. Locality is preserved: the wavepacket width evolves from 2.20 to 2.96 over $t = 1.5$, corresponding to a propagation speed $v \approx 0.5$ sites per time unit, below the causal bound set by the Laplacian spectrum.

The spectral purification $S_{\text{DNLS}}(t) \rightarrow 0$ and the thermodynamic second law $S_{\text{total}} \geq 0$ are complementary. The arrow of time in the early universe (decreasing quantum uncertainty during inflation) and the arrow of time in macroscopic thermodynamics (increasing disorder) point in the same direction because the same irreversible dissipation—coherence transferred from the system to the lattice reservoir—generates both simultaneously.

6 Testable predictions

A. Fixed-width attractor (FCC lattice.) Prepare a BEC in a 3D FCC optical lattice with attractive interactions. Inject a localized pulse. Predict: width $\rightarrow 2.96$ sites. Falsified if the width diverges or collapses to a single site.

B. Self-trapping threshold (norm = 4.) Vary atom number. Predict: abrupt transition at norm 4 between expansion and self-trapping. Falsified if the transition is gradual.

C. Dimensional dichotomy (1D vs. 3D.) Repeat in 1D tube and 3D FCC. Predict: 1D degrades; 3D stabilizes. Falsified if both dimensions yield the same behavior.

D. Ensemble purification ($S(t) \rightarrow 0$.) Run $M \geq 100$ repetitions, measure von Neumann entropy. Predict: $S(t) \rightarrow 0$, rank $\rightarrow 1$. Falsified if $S(t) > 0$ persists.

E. Spectral zeta functional equation. Verify $\xi(3/2-s) = \xi(s)$ numerically. Error $< 2\%$ at $s = 3/4$. Falsified if error diverges with N .

Existing experimental work supports the required components: reservoir-engineered DNLS dissipation [6], dissipative BEC purification [7], 3D discrete solitons [8], and continuous BEC [9].

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7 Conclusions

We have constructed a discrete spacetime model on an FCC lattice with zero free parameters that: (i) produces stable solitons from localized pulses, (ii) reveals a spectral attractor resolving the measurement problem in 3D, (iii) breaks time-reversal symmetry via irreversible purification ($S(t) \rightarrow 0$), (iv) predicts a self-trapping threshold (norm 4), and (v) connects to Epstein’s zeta function and the Riemann hypothesis. The collapse rate $\beta = (1 - \eta) + \varepsilon_0/2$ is derived from the FCC packing fraction, closing the model entirely. Five falsifiable experiments are proposed, realizable with existing cold-atom platforms.

References

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